

20/5/22 Deterministic finite automata (DFA)

- Q - a finite set of states
- Σ - a finite set of input symbols (alphabet)
- q_0 - a start state
- F - set of final states
- δ - a transition function, which is a mapping between $Q \times \Sigma \Rightarrow Q$

→ A DFA is defined by the 5-tuple:

$$\{Q, \Sigma, q_0, F, \delta\}$$

23/5/22 Example transition table

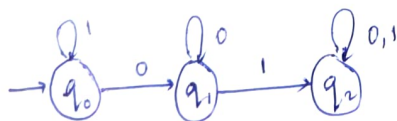
δ	0	1
→ q_0	q_1	q_0
q_1	q_1	q_2
* q_2	q_2	q_2

δ is represented either using state diagram or transition table.

Conventional form of writing

Start state: → q_0 or $\delta = \{q_0\}$

Final state: * q_2 or $F = \{q_2\}$



Final states
→ q_2

0111

→ q_2

0100

→ q_2

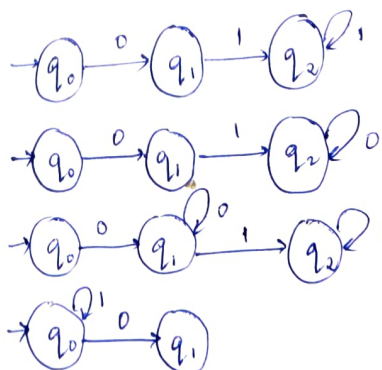
0011

→ q_2

1110

→ q_1

(not accepted)



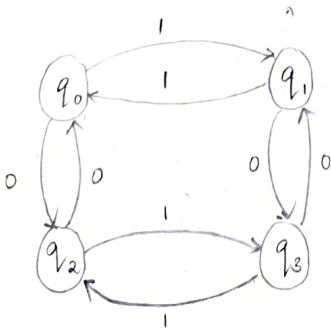
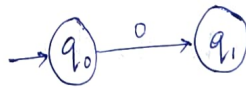
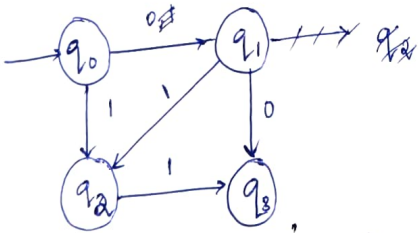
Clamping logic: waits for a "1" input and turns on forever. To avoid clamping on spurious noise, we'll design a DFA that waits for two consecutive 1's in a row before clamping on.

$L = \{ w \mid w \text{ is a bit string which contains the substring } 11 \}$

δ	0	1
$\rightarrow q_0$	q_0	q_1
q_1	q_0	q_2
$*q_2$	q_2	q_2

⑤ Build a DFA for,

$L = \{ w \mid w \text{ is a binary string that has even number of 1's and even number of 0's} \}$



$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

start state : $S = \{q_0\}$

$F = \{q_3\}$

δ	0	1
q_0	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_3
q_3	q_1	q_2

Q Design a DFA which accepts even numbers in binary form.

$$\rightarrow L = \{ \cancel{0}, 10, 100, 110, 1000, \dots \}$$

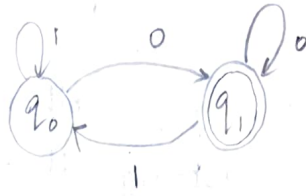
$$Q = \{q_0, q_1\}$$

$$S, P = \{q_0\}$$

$$F = \{q_1\}$$

$$\Sigma = \{0, 1\}$$

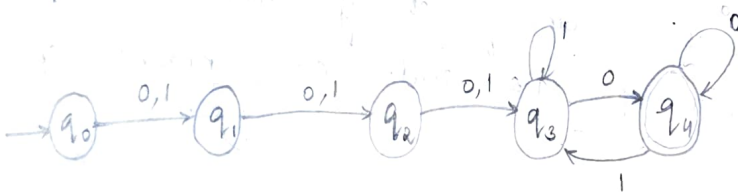
δ	0	1
q_0	q_1	q_0
q_1	q_1	q_0



Q Design a DFA which accepts binary string of minimum length 4 and is divisible by 2.

$$L = \{ 0000, 0010, 0100, 0110, 1000, 1010, 1100, 1110, 10000, \dots \}$$

$$\text{No. of states} : 4 + 1 = 5$$



$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$S = \{q_0\}$$

$$F = \{q_4\}$$

$$\Sigma = \{0, 1\}$$

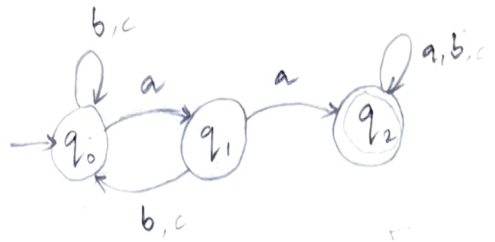
δ	0	1
q_0	q_1	q_1
q_1	q_2	q_2
q_2	q_3	q_3
q_3	q_4	q_3
q_4	q_4	q_3

8) Construct a DFA that accepts all strings having aa as a substring where the alphabet

$$\Sigma = \{a, b\}$$

→

δ	a	b
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_2



$$Q = \{q_0, q_1, q_2\}$$

$$S = \{q_0\}$$

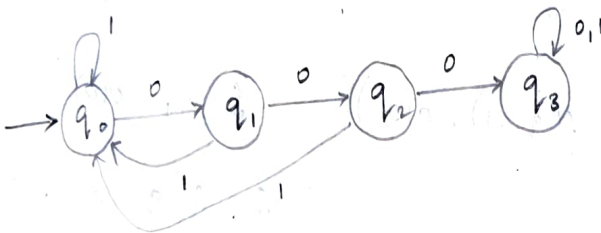
$$F = \{q_2\}$$

$$\Sigma = \{a, b\}$$

$$\text{if } \Sigma = \{a, b, c\},$$

9) Construct a DFA that accepts all binary strings with 3 consecutive zeroes.

$$\rightarrow L = \{000, 1000, 0001, \dots\}$$



δ	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_3	q_0
q_3	q_3	q_3

$$\Sigma = \{0, 1\}$$

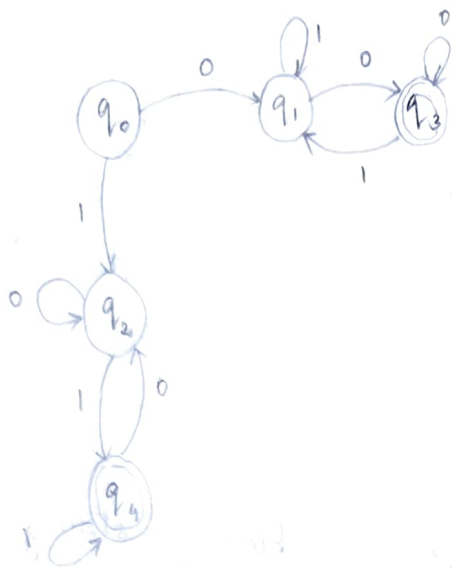
$$Q = \{q_0, q_1, q_2, q_3\}$$

$$S = \{q_0\}$$

$$F = \{q_3\}$$

Q Construct a DFA to accept all binary strings which start and end with same bit.

$$\rightarrow L = \{000, 010, 00, 11, 010100, \dots\}$$



δ	0	1
q_0	q_1	q_2
q_1	q_1	q_2
q_2	q_2	q_4
q_3	q_3	q_1
q_4	q_2	q_4

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$S = \{q_0\}$$

$$F = \{q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

Q Extending transition function:

for above machine,

$$\delta(q_0, 0) = q_1$$

$$\begin{aligned}
 \rightarrow \hat{\delta}(q_0, 0101) &= \hat{\delta}(\delta(q_0, 1), 0101) = \hat{\delta}(q_2, 0101) \\
 &= \hat{\delta}(\delta(q_2, 0), 101) = \hat{\delta}(q_4, 101) \\
 &= \hat{\delta}(\delta(q_4, 1), 01) \\
 &= \hat{\delta}(q_2, 01) = \hat{\delta}(\delta(q_2, 0), 1) \\
 &= \hat{\delta}(q_4, 1) = \hat{\delta}(\delta(q_4, 1), 1) \\
 &= \hat{\delta}(q_4, 11) = \hat{\delta}(q_2, 1) = \hat{\delta}(\delta(q_2, 1), 1) \\
 &= \hat{\delta}(q_4, 111) = \hat{\delta}(q_2, 111) = \hat{\delta}(\delta(q_2, 1), 11) \\
 &= \hat{\delta}(q_4, 1111) = \hat{\delta}(q_4, 1111) = q_4
 \end{aligned}$$

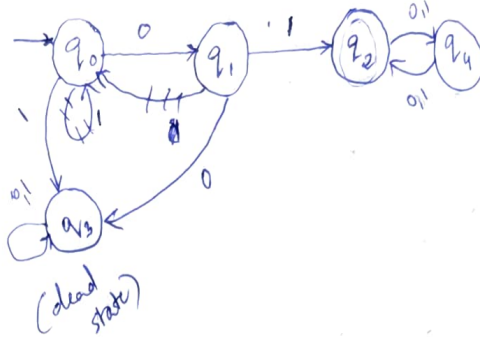
$$\rightarrow \delta(q_0, 0101) =$$

Q Design a DFA that accepts the language where

$L = \{ w \mid w \text{ is of even length and begins with 01 zero} \}$

$\Sigma = \{ 0, 1 \}$

$L \neq \epsilon$



δ	0	1
q_0	q_1	q_3
q_1	q_3	q_2
q_2	q_4	q_4
q_3	q_3	q_3
q_4	q_2	q_2

Non-deterministic Finite Automata (NFA)

→ Emptying a machine can exist in more than one state at the same time.

eg: $L = \{ \text{second position is 1 in binary string} \}$

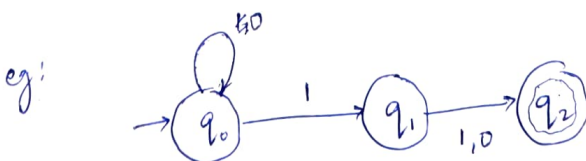
$L = \{ 01, 11, 010, 011, \dots \}$

⇒ machine can be in accepting state or intermediate state after reading second position 1.

NFA consists of:

Q, Σ, q_0, F

δ : a transition function, a mapping between $Q \times \Sigma \Rightarrow$ subset of Q . (can have multiple states of q_0, q_1)



δ	0	1
q_0	q_0	$\{q_0, q_1\}$
q_1	q_2	q_2
q_2		

$(q_0, 0) \xrightarrow{\quad} (q_1, 0)$
 $\downarrow q_2 \times$
 $(q_0, 1)$
 $\downarrow$
 $(q_0, 1) \xrightarrow{\quad} (q_1, 1)$
 $\downarrow q_2$
 $(q_0, 1) \xrightarrow{\quad} q_1 \times$
 $\downarrow$
 $(q_0, 1) \xrightarrow{\quad} (q_1, 0)$
 $\downarrow$
 $(q_0, 0)$
 $\downarrow$
 $q_0 \times$

```

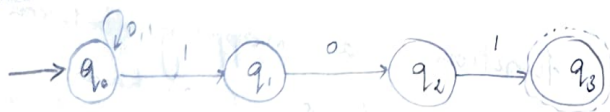
graph LR
    start(( )) --> q0((q0))
    q0 -- "0, 1" --> q0
    q0 -- "0" --> q1((q1))
    q1 -- "1" --> q2(((q2)))
    q2 -- "0, 1" --> q2
  
```

δ	0	1
q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_2\}$

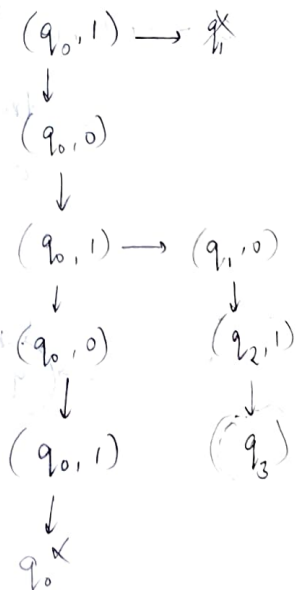
start state : q_0

$$\Sigma = \{0, 1\}$$
$$Q = \{q_0, q_1, q_2\}$$

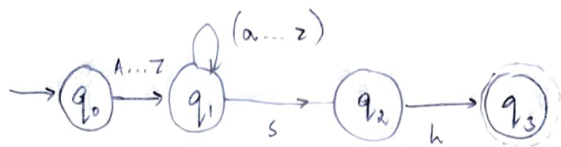
3. Construct an NFA that accepts all binary strings that end with 101.



δ	0	1
q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	ϕ
q_2	ϕ	$\{q_3\}$
q_3	ϕ	ϕ

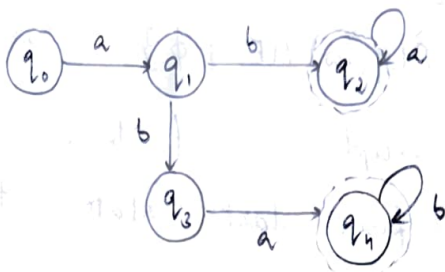


- $$\Sigma = \{a, b, \dots, z, A, B, \dots, Z\}$$

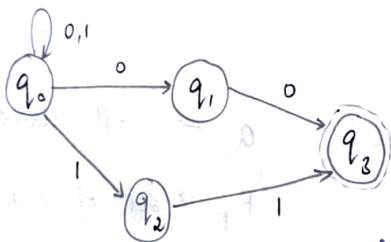


- $$L = \{ w \mid w \in abab^n \text{ or } aba^n \text{ where } n \geq 0 \}$$

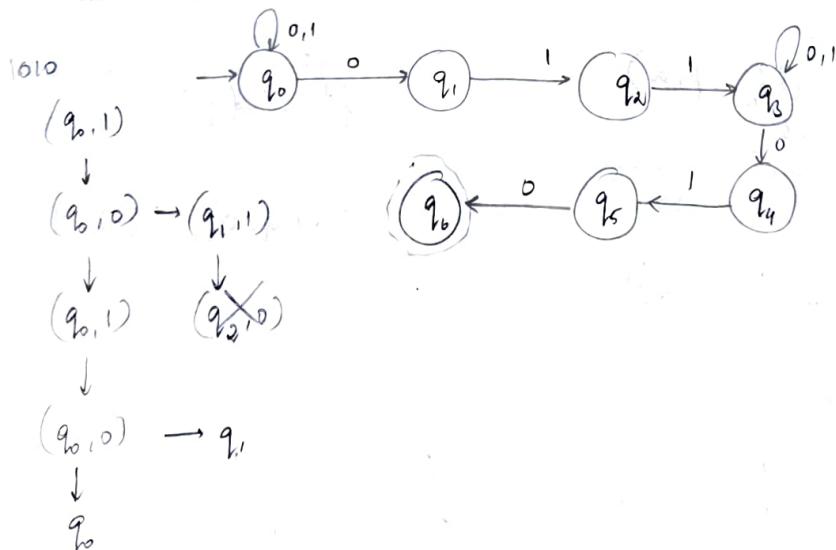
aba
abab
ababb
ab
abaa



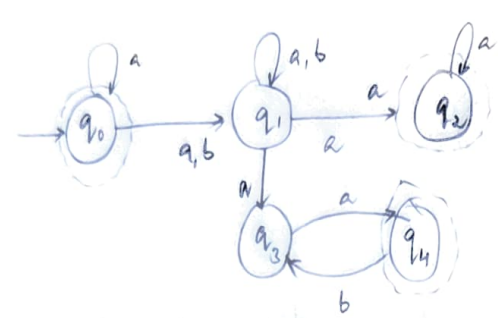
- 00, 11, 100, 000, 011, 111, ...



- 2) NFA that accepts a set of strings of 0's and 1's, ends with 010 and has 011 somewhere previously



8) Construct NFA that ends with a^* or $(ab)^*$
 0 or more $a, aa, ab, \dots abab, \dots$



Language of NFA

$$L(N) = \{w \mid \hat{\delta}(q_0, w) \cap F \neq \emptyset\}$$

As NFA accepts w if there exists at least one path from start state to an accepting state that is labeled by w .

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NFA to DFA Conversion

Subset Construction Method

$$N = (Q_N, \Sigma, q_0, F, \delta_N)$$

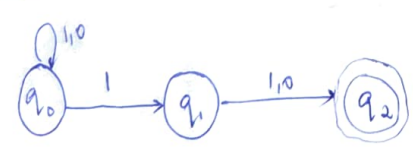
$$D = (Q_D, \Sigma, \{q_0\}, S, \delta_D)$$

$Q_D \rightarrow$ all subsets of Q_N

$F_D \rightarrow$ set of subsets s of Q_N s.t. $s \cap F_N \neq \emptyset$

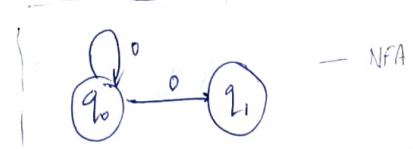
$$\delta_D(s, a) = \bigcup_{p \in s} \delta_N(p, a)$$

eg:



NFA $\rightarrow$ last symbol 1

1 Subsets $\rightarrow \delta(q_0, q_1, q_2)$



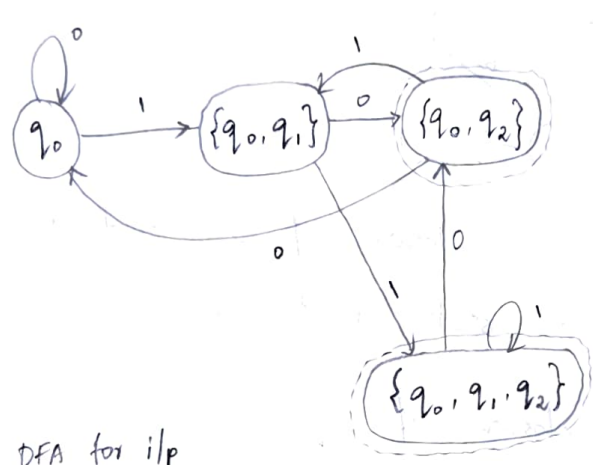
$s_0 \Rightarrow (q_0, 0) \rightarrow (q_0, q_1)$
 new state in DFA that

δ	0	1		
$\emptyset$	$\emptyset$	$\emptyset$	A	x
$\rightarrow \{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$	B	✓
$\{q_1\}$	$\{q_2\}$	$\{q_2\}$	C	x
* $\{q_2\}$	$\emptyset$	$\emptyset$	D	✓

δ	0	1
$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$ E ✓
$\{q_0, q_2\}$	$\{q_0\}$	$\{q_0, q_1\}$ F ✓
$\{q_1, q_2\}$	$\{q_2\}$	$\{q_2\}$ G ✗
$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$ H ✓

Discard all ^{states} that are not accessible from start state.

$\{B, E, F, H\}$ states are accepted.

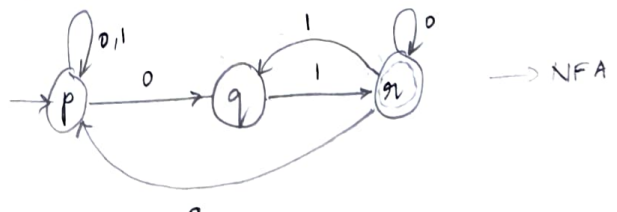


δ	0	1
$\{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$ already covered
$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$ next state to be considered
$\{q_0, q_2\}$	$\{q_0\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_0, q_1, q_2\}$

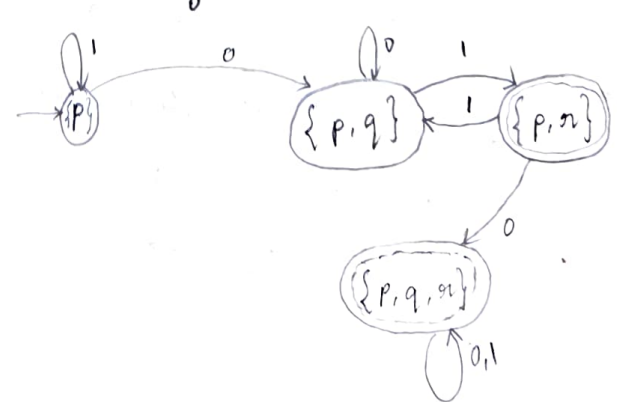
Trace DFA for i/p
00010 :

$\rightarrow (q_0, 0) \rightarrow (q_0, 0) \rightarrow (q_0, 1) \rightarrow (\{q_0, q_1\}, 0) \rightarrow \{q_0, q_2\}$

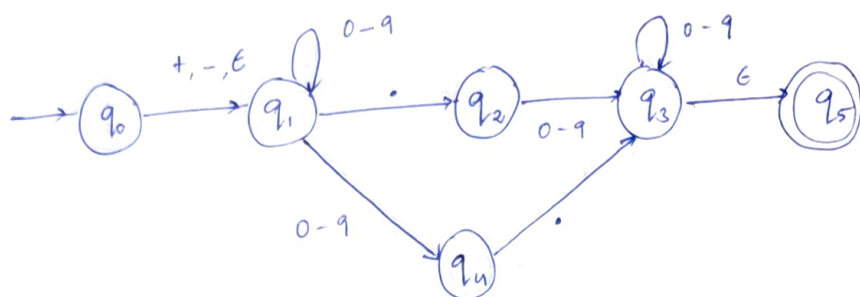
g) Convert following NFA to DFA.



	δ	0	1
$\times \rightarrow \phi$	ϕ	ϕ	ϕ
$\checkmark$	p	$\{p, q\}$	$\{p\}$
$\times$	q	ϕ	$\{r\}$
$\times$	* r	$\{p, r\}$	$\{q\}$
$\checkmark$	$\{p, q\}$	$\{p, q\}$	$\{p, r\}$
$\checkmark$	$\{p, r\}$	$\{p, q, r\}$	$\{p, q\}$
$\times$	$\{q, r\}$	$\{p, r\}$	$\{q, r\}$
$\checkmark$	$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$

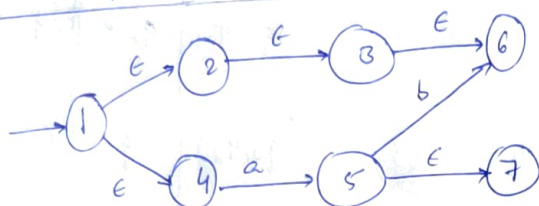


2/6/22 Finite Automata with Epsilon Transitions



- 2. ✓
- 0.1 ✓
- .1 ✓

ε-closure of a state



Defⁿ: If state p is in ϵ -closure of q and there is a transition from state p to state r , labeled ϵ then r is ϵ -closure of q .

ϵ -closure of $(1) = \{1, 2, 4, 3, 6\}$

ϵ -closure of $(2) = \{2, 3, 6\}$

ϵ -closure of $(3) = \{3, 6\}$

ϵ closure of $(4) = \{4\}$

ϵ closure of $(5) = \{7\}$

ϵ closure of $(6) = \{6\}$

ϵ closure of $(7) = \{7\}$

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* In DFA, we do not have any transitions with ϵ

* Step 1: ϵ -closure (q_0) is the start state of DFA.

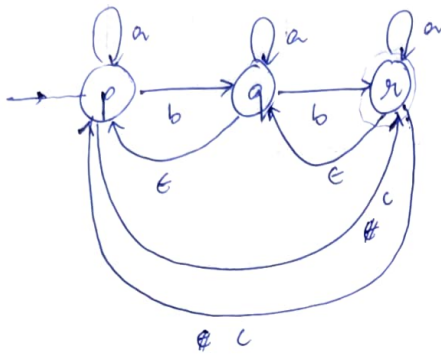
i.e if state 1 is start state, ϵ -closures of 1,
i.e $\{1, 2, 4, 3, 6\}$ can also be start state in DFA.

Step 2: Compute the transitions for DFA. Let $\{q_i, q_j, \dots, q_k\}$ is the state of DFA. Then transitions from this state on 'a' i.e. $\delta_d(\{q_i, q_j, \dots, q_k\}, a)$ is computed

as $\delta_e(\{q_i, q_j, \dots, q_k\}, a) = \{p_1, p_2, \dots, p_m\}$.

Then take ϵ -closure $(\{p_1, p_2, \dots, p_m\})$

Thus $\delta_d(\{q_i, q_j, \dots, q_k\}, a) = \epsilon\text{-closure}(\delta_e(\{q_i, q_j, \dots, q_k\}, a))$



DFA $\{Q_D, \Sigma, \delta_D, q_D, F_D\}$

Step 1:

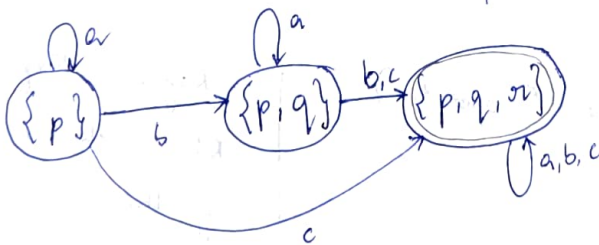
ϵ -closure of $(p) = \{p, \epsilon\}$

ϵ -closure of $(q) = \{q, p, \epsilon\}$

ϵ -closure of $(r) = \{r, p, q\}$

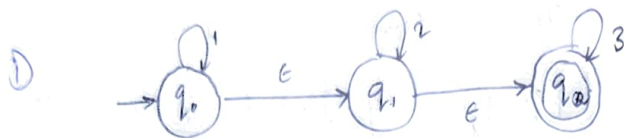
Step 2:

δ_D	a	b	c
$\{p\}$	$\{p\}$	$\{q\}$	$\{r, p, q\}$
$\{p, q\}$	$\{p, q\}$	$\{p, q, r\}$	$\{p, q, r\}$
$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$



Language : $\{c, bb, bc, abb, abc, abac, \dots\}$

⑤ Convert the following ϵ -NFA to DFA.



ϵ -closure of $q_0 \rightarrow \{q_0, q_1, q_2\}$

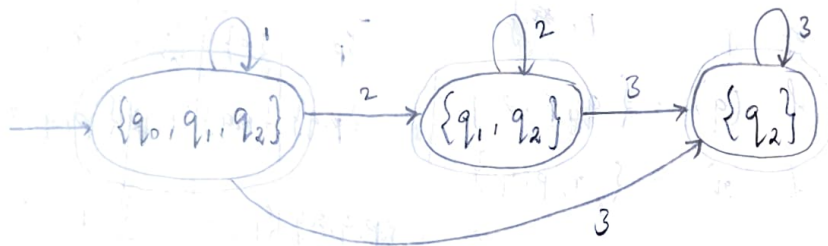
ϵ -closure of $q_1 \rightarrow \{q_1, q_2\}$

ϵ -closure of $q_2 \rightarrow \{q_2\}$

δ	1	2	3
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
$\{q_1, q_2\}$	$\emptyset$	$\{q_1, q_2\}$	$\{q_2\}$
$\{q_2\}$	$\emptyset$	$\emptyset$	$\{q_2\}$

eg: Transaction
when not entered,
Resend OTP
is an example for
 ϵ

δ	1	2	3
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
$\{q_1, q_2\}$	$\emptyset$	$\{q_1, q_2\}$	$\{q_2\}$
$\{q_2\}$	$\emptyset$	$\emptyset$	$\{q_2\}$



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Kleene Closure (the $*$ operator)

let $L = \{1, 00\}$

$L^0 = \{\epsilon\}$

$L^1 = \{1, 00\}$

$L^2 = \{11, 0000, 100, 001\}$

Regular Expressions

RE which is accepted by the automata

eg: $01^* + 10^*$

$L(01) = \{01\}$

$L(01+0) = \{01, 0\}$

$L(0(1+0)) = \{01, 00\}$

$L(0^*) = \{\epsilon, 0, 00, 000, \dots\}$

Building Regular Expressions:

• let E be a regular expression and the language represented by E is $L(E)$

then,

$$L(E) = E$$

$$L(E + F) = L(E) \cup L(F)$$

$$L(EF) = L(E) L(F)$$

$$L(E^*) = (L(E))^*$$

① $a(aa^*) \Rightarrow L = \{a, aa, aaa, \dots\}$

② $(a+b)^* \Rightarrow L = \{\epsilon, a, b, ab, aa, aaa, bb, baa, \dots\}$

③ $01^* + 10^* \Rightarrow L = \{0110, 01, 0111, 10, 01, \dots\}$

Precedence of Operators

$*$ $\cdot$ $+$
Highest $\longrightarrow$ lowest

$$(a+b)^* = (a^* + b^*)$$

a^* b^*

① $a(aa^*) \Rightarrow L = \{a, aa, aaa, \dots\}$

② $(a+b)^* \Rightarrow$

③ $L((0+10)^*(\epsilon+1)) =$ all strings of 0's & 1's without two consecutive 1's.

$$= \{\epsilon, 1, 0, 10, \dots\}$$

④ $L((0+1)^* 101 (0+1)^*) =$ all strings of 0's and 1's having 101 as a substring

0110101

$1^* 0^*$
 $(10)^*$
 ϵ
100
10
11
10100
1111000

⑧ $L((0+1)^*1(0+1)^*0(0+1)^*1(0+1)^*)$ = all strings of 0's and 1's having 101 as a subsequence.
 $= \{ 101, 0101, 1101, 1001, 11001, 11101, \dots \}$

⑨ $L(1^*(1^*01^*01^*01^*)^*1^*)$ = all strings of 0's and 1's having a number of 0's that is a multiple of 3.
 $= \{ \epsilon, 000, \dots \}$

1. Set of string of a's and b's ending with "abb"

$\rightarrow \cancel{a^*b} \cancel{a^*b} \cancel{abb} (ba)^*(ab)^*(abb)^*abb$

2. Set of string of a's and b's starting with "ab"

$\rightarrow ab(ab)^*$

3. Set of string of a's and b's having substring "aa"

$\rightarrow (ab)^*aa(ab)^*$

4. Set of strings consisting of atleast one "a" followed by string consisting of atleast one "b"

$\rightarrow aa^*b^*b$

5. Set of strings that contain exactly 3 a's.

$\rightarrow b^*ab^*ab^*ab^*$

vi) Strings of a's and b's containing no more than 3 a's. 9/6/22

$$\rightarrow (b^*) + (b^*ab^*) + (b^*ab^*ab^*) + (b^*ab^*ab^*ab^*)$$

(or) $b^*(a+\epsilon)b^*(a+\epsilon)b^*(a+\epsilon)b^*$

vii) Ending with either "ab" or "ba"

$$(a+b)^*a^*b^*(ab+ba) \quad (a+b)^*(ab+ba)^*(ab+ba)$$

~~aba (ab+ba)~~
abaab

viii) Starting with "a" and ending with "b"

$$\rightarrow a(a+b)^*(ab+ba)^*b$$

ix) Whose lengths are multiple of 3

$$\rightarrow ((a+b)(a+b)(a+b))^*$$

x) Fourth symbol from right end is "a" &
fifth symbol from right end is b

$$\rightarrow (a+b)^*ba(a+b)(a+b)(a+b)$$

→ (can be implemented using NFA, since we cannot determine when the string terminates)

⑧ language of

$$(0+1)^* 0 (0+1)^* 0 (0+1)^*$$

→ 00 as subsequence

✓ (or) set of all strings containing atleast 2 0's.

⑨ all string that begin & end with either 0 or 1 $\times$ [not accepting 11]

⑩ The regular expression set as

$0^* (10^*)^*$ denotes same

$$(10)^*$$

$$1^* 0^*$$

A) $(1^* 0)^* 1^*$

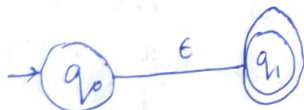
B) $0 + (0+10)^*$

C) $(0+1)^* 10 (0+1)^*$

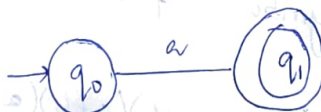
D) none of these



$L \rightarrow \text{NULL SET} = \phi$



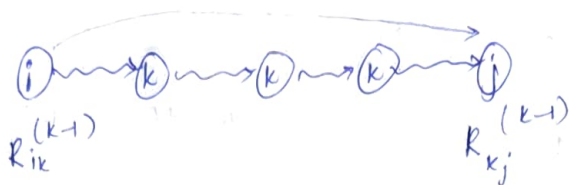
$L = \epsilon$



$L = \{a\}$

13/6/22

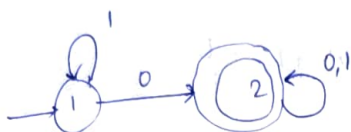
FA to RE



$$R_{ij}^k = R_{ij}^{k-1} + R_{ik}^{k-1} (R_{kk}^{k-1})^* R_{kj}^{k-1}$$

8

8) Convert following DFA to RE



$$R_{12}^{(2)} = R_{12}^{(1)} + R_{11}^{(1)} (R_{22}^{(1)})^* R_{22}^{(1)}$$

$$= R_{12}^0 + R_{11}^0 (R_{22}^0)^* R_{22}^0$$

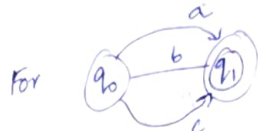
$$= \left(R_{12}^0 + R_{11}^0 (R_{22}^0)^* R_{22}^0 \right)$$

$$R_{12}^1 = 0 + 1(1)^* 0 = 1^* 0$$

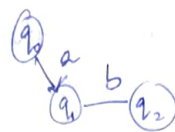
$$R_{22}^1 = (0 + 1 + \epsilon) + \phi (1 + \epsilon)^* 0 = 0 + 1 + \epsilon$$

$$R_{12}^2 = 1^* 0 + (1^* 0) (0 + 1 + \epsilon)^* (0 + 1 + \epsilon)$$

$$= 1^* 0 (0 + 1 + \epsilon)^*$$



$a+b+c$



$a.b$

concatenation requires intermediate state

$$R_{12}^1 = R_{12}^0 + R_{11}^0 (R_{22}^0)^* R_{22}^0$$

$$R_{22}^{(1)} = R_{22}^0 + R_{21}^0 (R_{11}^0)^* R_{12}^0$$

$$R_{11}^0 = 0$$

$$R_{11}^0 = 1 + \epsilon$$

$$R_{21}^0 = \phi$$

$$R_{22}^0 = \{0 + 1\} + \epsilon$$

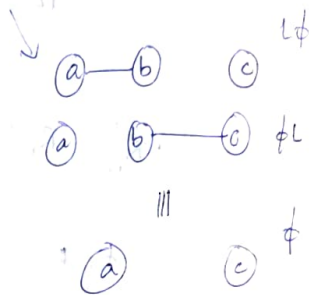
$$R_{11}^1 = 1^* \text{ (not req)}$$

$$= \epsilon + 1 + (\epsilon + 1)(\epsilon + 1)^* = (\epsilon + 1)^*$$

Algebra for RE's

- Commutative law for union, $L + M = M + L$
- Associative law for union, $(L + M) + N = L + (M + N)$
- Associative law for concatenation, $(LM)N = L(MN)$
- Commutative law is not applicable for concatenation.
 $LM \neq ML$

- Identity for union, $L + \phi = \phi + L = L$
- $LE = EL = L$
- $\phi L = L\phi = \phi$ (annihilator for concatenation)
- $L(M + N) = LM + LN$
- $(M + N)L = ML + NL$
- Idempotent law, $L + L = L$
- $L.L = LL$ (cannot be only L)



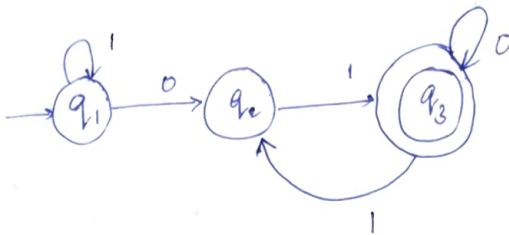
Rules Involving Closure :

- $(L^*)^* = L^*$
- $\epsilon^* = \epsilon$
- $L^* = L^* \epsilon + \epsilon$
- $\phi^* = \epsilon$

Identities and Annihilators

- $R + \phi = R$, ϕ is identity for $+$
- $\epsilon R = R\epsilon = R$, ϵ is identity for concatenation
- ϕ is the annihilator for concatenation
 $\phi R = R\phi = \phi$

⑧ Convert the DFA to RE



Let $q_1 \rightarrow 1$
 $q_2 \rightarrow 2$
 $q_3 \rightarrow 3$

$$R_{ij}^k = R_{ij}^{k-1} + R_{ik}^{k-1} (R_{kk}^{k-1})^* R_{kj}^{k-1}$$

$$R_{13}^{(3)} = R_{13}^2 + R_{13}^2 (R_{33}^2)^* R_{33}^2$$

$$\begin{aligned}
 R_{13}^2 &= R_{13}^1 + R_{12}^1 (R_{22}^1)^* R_{23}^1 = \phi + R_{12}^1 (1) = 1 \\
 R_{33}^2 &= R_{33}^1 + R_{32}^1 (R_{22}^1)^* R_{23}^1 = \phi + R_{32}^1 (1) = \phi + 1 \\
 R_{13}^1 &= R_{13}^0 + R_{11}^0 (R_{11}^0)^* R_{13}^0 = \phi + (1+\epsilon)(1+\epsilon)\phi = \phi \\
 R_{22}^1 &= R_{22}^0 + R_{21}^0 (R_{11}^0)^* R_{12}^0 = \epsilon + \phi(1+\epsilon)^0 = \epsilon \\
 R_{23}^1 &= R_{23}^0 + R_{21}^0 (R_{11}^0)^* R_{13}^0 = 1 + \phi(1+\epsilon)^0 = 1 \\
 R_{12}^1 &= R_{12}^0 + R_{11}^0 (R_{11}^0)^* R_{12}^0 = 0 + (1+\epsilon)(1+\epsilon)^0 = 0 + (1+\epsilon) \\
 R_{33}^1 &= R_{33}^0 + R_{31}^0 (R_{11}^0)^* R_{13}^0 = 0 + \epsilon + \phi(1+\epsilon)^0 = 0 + \epsilon \\
 R_{32}^1 &= R_{32}^0 + R_{31}^0 (R_{11}^0)^* R_{12}^0 = 1 + \phi(1+\epsilon)^0 = 1 + \phi \\
 R_{13}^0 &= \phi \\
 R_{11}^0 &= 1 + \epsilon \\
 R_{22}^0 &= \phi + \epsilon = \epsilon \\
 R_{21}^0 &= \phi \\
 R_{12}^0 &= 0 \\
 R_{33}^0 &= 0 + \epsilon \\
 R_{32}^0 &= 1 \\
 R_{31}^0 &= \phi \\
 R_{23}^0 &= 1
 \end{aligned}$$

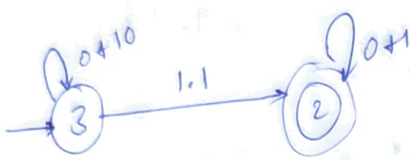
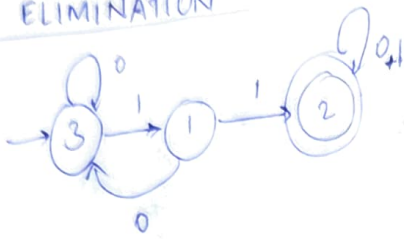
$\Rightarrow$ For n states, n^2 eq^s need to be calculated.

$$\begin{aligned}
 R_{13}^{(3)} &= R_{13}^{(2)} + R_{13}^{(2)} (R_{33}^{(2)})^* R_{33}^{(2)} \\
 &= \phi + R_{12}^1 \cdot 1 + \phi R_{12}^1 (0 + \epsilon + 1)^* (0 + \epsilon + 1) \\
 &= 0 + (1 + \epsilon)(1 + \epsilon)^0 \cdot 1 + 0 + (1 + \epsilon)(1 + \epsilon)^0 (0 + \epsilon + 1)(0 + \epsilon + 1) \\
 &= (1 + \epsilon)^0 \cdot 1 + (1 + \epsilon)^0 \cdot (0 + \epsilon + 1)^* (0 + \epsilon + 1) \\
 &= 1^* 0 \cdot 1 + 1^* 0 (0 + 1)^*
 \end{aligned}$$

FA: 1*01(0+11)*

STATE ELIMINATION

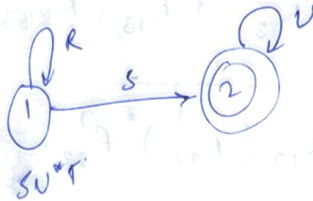
1 ①



$$\Rightarrow (0+10)^* 1 (0+1)^*$$

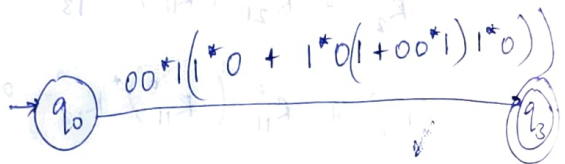
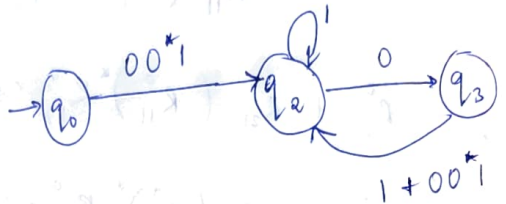
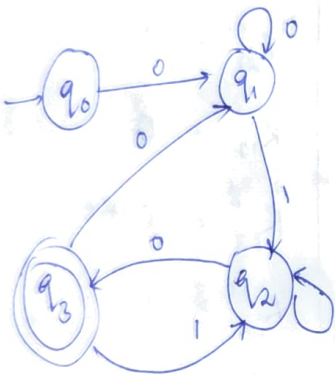
WITH LOOPS

2 ②



$$\Rightarrow (R^* + SU^*T)^* SU^*$$

3 ③

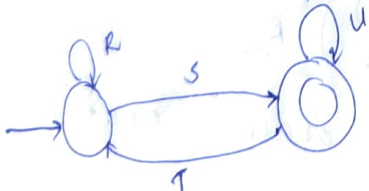


$$= 00^*1(1^*0 + 1^*0(1+00^*1)1^*0)$$

$$= 00^*11^*0((1+00^*1)1^*0)^*$$

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State Elimination method,



2 state automata

$$(R + SU^*T)^* SU^*$$

$$R^*(R + SU^*T)^* SU^*$$

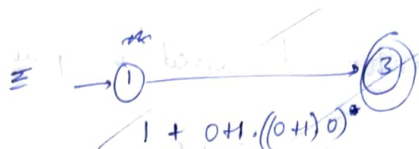
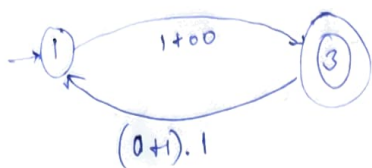
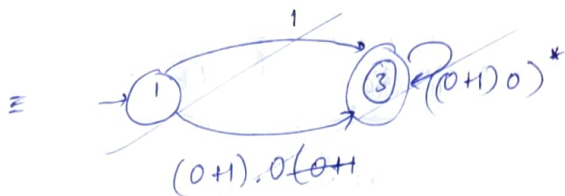
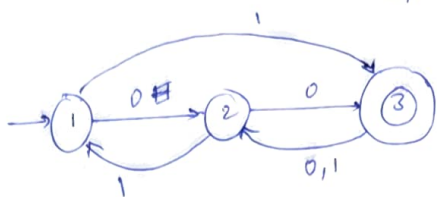
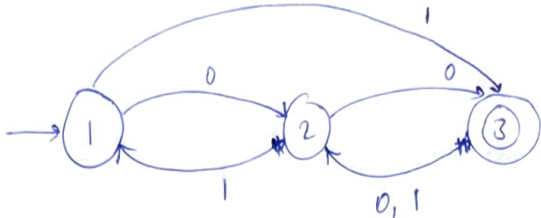
$$(R + R^*SU^*T)^* SU^*$$



$$\equiv R^*$$

1 state automata

Q)



$$(01)^* + ((00+1)^* ((0+1)0)^* (0+1)1)^*$$

$$(00+1) ((0+1)0)^*$$

$$(R+SU^*T) (R+SU^*T)$$

20/6/22

Applications of RE ;

① In UNIX,

→ operator | used in place of +

→ $R?$ $\Rightarrow$ $\epsilon + R$

→ $R\{n\} \Rightarrow$ R concatenated n times.

→ $\cdot \Rightarrow$ any character

ie a. $\Rightarrow$ ab (or) ac ...

→ lexical Analyser (scans until a meaningful block is obtained)

Statement of pumping lemma:

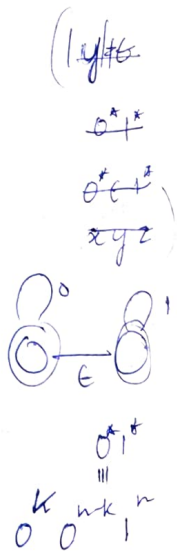
For every regular language L,

1. $|xyz| \leq n$

2. $|y| > 0$

3. For all $k \geq 0$, xy^kz is in L

$y \neq \epsilon$



⑧ 0^n where n is a prime no.

$$\rightarrow 0^n = xyz = \cancel{0^i 0^j 0^k} = 0^i (0^j)^k 0^{n-i-j}$$

$$= \left(\begin{matrix} 2+i & n-2-i \\ 0 & (6)0 \end{matrix} \right)$$

$$j+k + n - j = (k+1)j + n$$

$\Rightarrow k = \text{variable}$

For $k=1 \Rightarrow n$ ✓

For $k=2, \Rightarrow j+n$??

For $k=n+1 \Rightarrow n(j+1) \Rightarrow \text{composite}$

$i + (j+k) + n - i - j$
has to be a
prime no.

$\Rightarrow$ not a regular expression

⑨ Prove for $n = \text{perfect square}$, 0^n

$$\rightarrow 0^n = xyz = 0^i (0^j)^k 0^{n-i-j}$$

$\Rightarrow (k-1)j + n$ has to
be perfect square

(For $k=n+1, \Rightarrow n(j+1)$)

$$L = \{0, 0000, 0000000000 \dots\}$$

$$w = |xyz| \geq n$$

$$|y| \neq 0$$

$$|xy| \leq n$$

$$1 \leq y \leq n$$

$$|xyz| = 2$$

$$|xyz| + |y|$$

$$= n^2 + 1 \dots n^2 + n$$

for min $|y|$

$\neq$ perfect square

$((k-1)j + n)^{1/2}$ must be
an integer

$n=4,$	$5j$	$j < n$
$n=9,$	$10j$	
$n=16,$	$17j$	
$n=25,$	$26j$	

$$n^2 + n < (n+1)^2 \Rightarrow \text{floating point root}$$

sum of digits of a
perfect square $\rightarrow$ either
of 4, 7, 9

⑩ Prove for $n = \text{perfect cube}$, 0^n is a regular language

$$\rightarrow 0^n = xyz = 0^i (0^j)^k 0^{n-i-j}$$

$\Rightarrow (k-1)j + n$ has to be
a perfect cube.

$$w = |xyz| \geq n$$

$$\text{for } i \geq 2, xy^2z = |xyz| + |y|$$

$$\Rightarrow n^3 + 1 \dots n^3 + n$$

$$< (n^3 + 1)^3$$

8) Prove that $0^n 1^m$ is a regular language where $n \leq m$.

$$\rightarrow 0^n 1^m = (x^k y z) \Rightarrow 0^i (0^{n-i})^k 1^m \quad n+m \leq 2m$$

For $k=2$,

$$\Rightarrow \text{---} + 2n \neq m$$

$$\begin{aligned} i + (n-i) + m &\leq 2m \\ i + n - i + m &\leq 2m \\ \Rightarrow (1-k)i + kn &\leq m \end{aligned}$$

(or)

$$0^n 1^m = 0^{n-i} (0^i)^k 1^m$$

$$n-i + ki + m \leq 2m$$

$$n + (k-1)i \leq m$$

$$1 \leq i$$

For any finite value $k > 1$, $n + \text{---} \neq m$

(contradiction to $n \leq m$)

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Closure Property of Regular Language

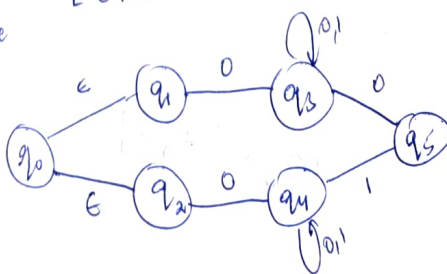
1. Union '+'
2. Klein closure '*'
3. Concatenation '.'
4. Complement

$\Rightarrow$ Let L & M be regular language.

R & $S \rightarrow$ regular exp. representing L & M .

$R+S \Rightarrow$ language $L \cup M$

eg: $L_1: 0(0+1)^*0$
 $L_2: 0(0+1)^*1$

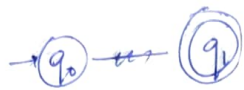


Closure under Intersection

Statement: If L & M are R. L, then so is $L \cap M$

- Construct M_3 , product automaton of A & B
↓
cartesian product of states

M_1



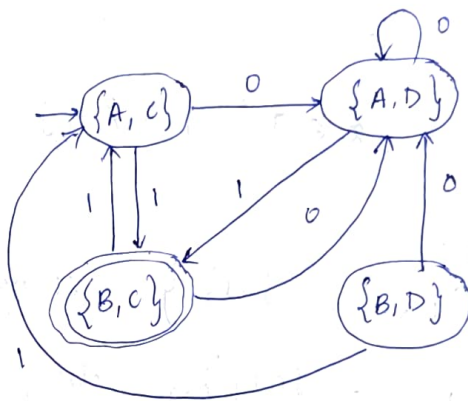
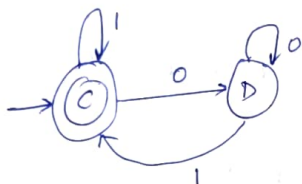
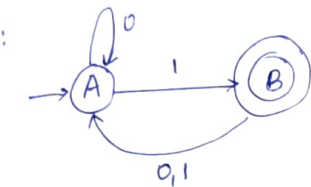
i.e. $q_0q_2, q_0q_3, q_1q_2, q_1q_3 \dots$

$q_0q_2 \rightarrow$ would be starting state

$q_1q_3 \rightarrow$ accepting state

$q_0q_3, q_1q_2 \rightarrow$ intermediate states

eg:



$$L_1 = \{1, 01, 101, 111, \dots\}$$

$$L_2 = \{\epsilon, 1, 01, 101, 111, \dots\}$$

$$L_1 \cap L_2 = \{1, 01, 101, \dots\}$$

Complement

$$L' = \Sigma^* - L$$

Σ^* is regular

Closure under Intersection

example:

$L_1 = \{w, w \in (0+1)^* \text{ and no. of 0's is even}\}$

$L_2 = \{w, w \in (0+1)^* \text{ and no. of 1's is even}\}$

$L_1 = \{\epsilon, 00, 0011, 010, 001010, \dots\}$

$L_2 = \{\epsilon, 11, 101, 101011, \dots\}$

For L_1 , $M_1 = (Q_1, \Sigma, \delta_1, q_0^1, F_1)$

For L_2 , $M_2 = (Q_2, \Sigma, \delta_2, q_0^2, F_2)$

Closure under Complementation

$$\bar{L} = \{x \in \Sigma^* \mid x \notin L\}$$

example,

$L = \{0x \mid x \in \{0,1\}^*\} = \text{strings that start with 0}$

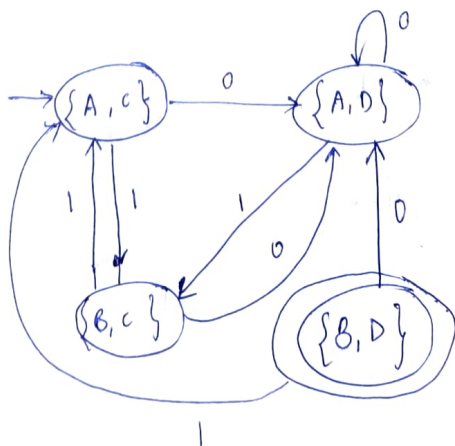
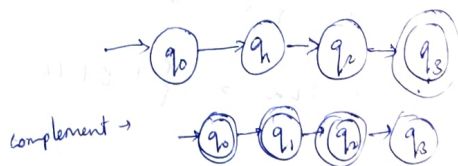
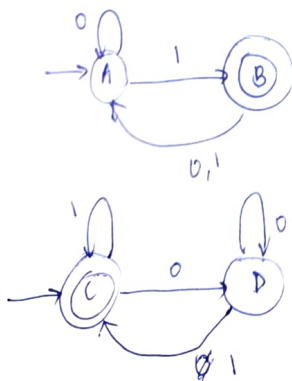
$\bar{L} = \{1x \mid x \in \{0,1\}^*\} \cup \{\epsilon\} = \text{strings that start with 1}$

Reverse accepting & non-accepting states

27/6/22

Difference :

$$A - B = A \cap \bar{B}$$



For difference
 $A - B$ product of
 Accepting state of A &
 non-accepting state of B
 $\Rightarrow$ accepting state of resultant

→ Here $\{B, D\}$ is not accepting any
 i/p $\Rightarrow A$ is a subset of B .

30/6/22

Reversal: of RE

Let $E = 01^* + 10^*$

Reversal $\rightarrow$

$$L = \{011, 100, 0111, 1000, 0, 1, \dots\}$$

$$\Rightarrow \text{Reversal: } 1^*0 + 0^*1 \quad \checkmark$$

$$[\because E^R = (01^*)^R + (10^*)^R]$$

Reversal of a RE gives another RE

$\Rightarrow$ Reversal RE is closed under reversal.

Homomorphism:

Suppose $0 \rightarrow ab$

$1 \rightarrow \epsilon$

$$01^* + 10^* \Rightarrow ab\epsilon^* + \epsilon(ab)^*$$

$$\Rightarrow ab + (ab)^* = (ab)^*$$

Inverse Homomorphism:

$$h^{-1}(L) = \{w \mid h(w) \text{ is in } L\}$$

$$L = \{abab, baba\}$$

$$h^{-1}(L) = L(1^*01^*01^*)$$

abab
00,

baba
 $0^R 0^R$

$0^R 20$
ab,
ba

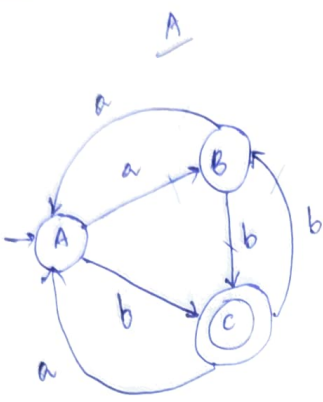
$\rightarrow A \cdot \bar{B}$
 $\rightarrow A \cup (\overline{A \cap B})$
 $\rightarrow A \cap \bar{B} \quad \checkmark$



$$11(.)^*11 \neq$$

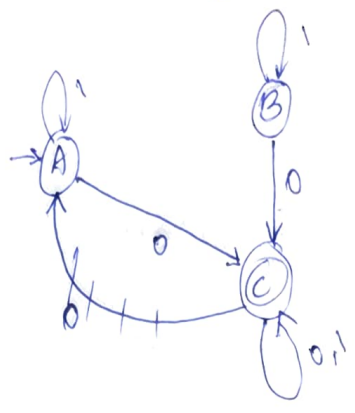
011111...
100...

1/7/22



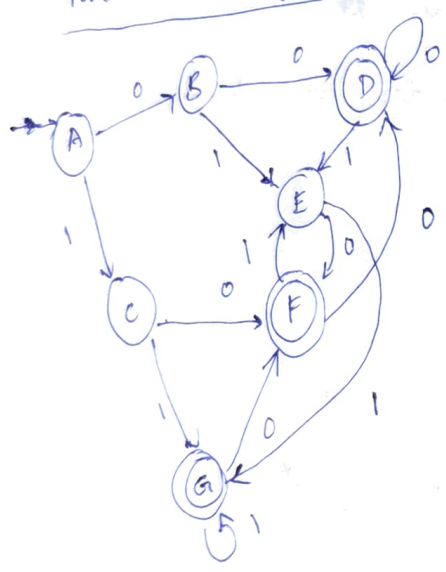
$h(0) = ab$
 $h(1) = \epsilon$

$$\delta_B(q, a) = \delta_A(q, h(a))$$



4/7/22

Table filling algorithm to minimize DFA



B	X					
C	X	X				
D	X	X	X			
E	X	X		X		
F	X	X	X		X	
G	X	X	X	X	X	X
	A	B	C	D	E	F

D, F, G belong to one set (accepting)
 others $\rightarrow$ (another set) (non-accepting)
 Non-matching sets $\Rightarrow \times$

- $\delta(A, 0) = B$
- $\delta(B, 0) = D$
- $\delta(B, 1) = E$
- $\delta(E, 0) = F$
- $\delta(D, 0) = D$
- $\delta(F, 0) = D$
- $\delta(G, 0) = F$
- $\delta(A, 1) = C$
- $\delta(C, 1) = G$
- $\delta(E, 1) = G$
- $\delta(D, 1) = E$
- $\delta(F, 1) = E$
- $\delta(G, 1) = G$

δ
 (E) $\neq$ (F) $\times$ removed

11/7/22

Context-free Grammar and Language :

eg: CFG for $\{0^n 1^n \mid n \geq 1\}$

neg: 0000011111

- i) $S \rightarrow 01$
- ii) $S \rightarrow 0S1$

0S1
00S11
000S111
0000S1111
0000011111 } using ii)
} using i)

CFG

Context-free Grammar is a 4-tuple $G = (V, T, P, S)$

V - A finite set of variables or non-terminals

T - A finite set of terminals ($V \cap T$ do not intersect i.e. do not use same symbols)

P - A finite set of productions

S -

example: CFG for $\{0^k 1^k \mid k \geq 0\}$:

$G = \{\{S\}, \{0, 1\}, P, S\}$

- P:
- i) $S \rightarrow 0S1$ or just $S \rightarrow 0S1 \mid \epsilon$
 - ii) $S \rightarrow \epsilon$

$L = \{a^n b^n \mid n \geq 0\}$

$G = \{\{S\}, \{a, b\}, P, S\}$

- P:
- i) $S \rightarrow aSb$

- ii) $S \rightarrow a$

9 Password for , starting with A-Z , any no. of numerals, combinⁿ of uppercase-lowercase, + , - , * (anywhere)

12/2/22 PALINDROME
00011000

OPO

using OPO

00POO

using OPO

000POOO

using IPI

0001P1000

using $P \rightarrow \epsilon$

00011000

$G_{pal} = (\{P\}, \{0,1\}, A, P, \{ \})$

Production rules:

$P \rightarrow \epsilon$

$P \rightarrow 0$

$P \rightarrow 1$

$P \rightarrow OPO$

$P \rightarrow IPI$

1. $\{w \in \{0,1\}^* \mid w \text{ contains at least 3 1's}\}$

$\rightarrow G = (\{S\}, \{0,1\}, P, S)$

Production rules,

i) $S \rightarrow SPS$

ii) $S \rightarrow OSO$

iii) $S \rightarrow ISI$

iv) $S \rightarrow OSI$

v) $S \rightarrow ISO$

vi) $S \rightarrow ISE$

vii) $S \rightarrow OS, SO$

i) ~~$S \rightarrow SPS$~~

ii) ~~$S \rightarrow \epsilon$~~

ii) ~~$S \rightarrow OS \mid IS \mid \epsilon$~~

$S \rightarrow XIXIXIX$

$X \rightarrow OX \mid IX \mid \epsilon$

$G = \{V, \Sigma, R, S\}$ with set $V = \{S, X\}$

where S is start variable
set of terminals $\Sigma = \{0,1\}$

- ① $w \in \{0,1\}^* \mid w \in w^R \text{ and } |w| \text{ is even} \}$

$$G = (\{P\}, \{0,1\}, A, P)$$

$$P \rightarrow \epsilon$$

$$P \rightarrow 0P0$$

$$P \rightarrow 1P1$$

- ③ $\{w \in \{0,1\}^* \mid \text{the length of } w \text{ is odd and the middle symbol is } 0\}$

$$G = (\{S, X\}, \{0,1\}, A, S)$$

$$S \rightarrow X0X$$

$$X \rightarrow 1X, 0X, \epsilon$$

$$1X01X$$

$$S \rightarrow SOS, \epsilon, X$$

$$X \rightarrow 1X, 0X$$

- ④ $\{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } i=k\}$

$$(S \rightarrow (Xab \text{ or } Xac) \rightarrow XaYb \text{ or } XaZe)$$

$$X \rightarrow a$$

$$Y \rightarrow b$$

$$Z \rightarrow c$$

$$S \rightarrow XaYZ$$

- ③ $\{w \in \{0,1\}^* \mid \text{the length of } w \text{ is odd \& the middle symbol is } 0\}$

$$S \rightarrow 0S0 \mid 1S1 \mid 0S1 \mid 1S0$$

$$S \rightarrow 0$$

$$G = (\{S\}, \{0,1\}, P, S)$$

$$1) \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } i=2k\}$$

$$G = \{V, \Sigma, R, S\} \quad V = \{S, W, X, Y, Z\}$$

$$S \rightarrow XY \mid W$$

$$X \rightarrow aXb \mid \epsilon$$

$$Y \rightarrow cY \mid \epsilon$$

$$W \rightarrow aWc \mid Z$$

$$Z \rightarrow bZ \mid \epsilon$$

$$2) \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i+j=k\}$$

$$S \rightarrow X \mid Y \mid \epsilon$$

$$X \rightarrow aWc, \quad Y \rightarrow Zb^Wc$$

$$W \rightarrow bc \mid ac, \quad Z \rightarrow a$$

$$Y \rightarrow bWc$$

~~$$S \rightarrow XW \mid YZ \mid \epsilon$$~~

~~$$X \rightarrow aXb \mid \epsilon$$~~

~~$$Y \rightarrow Zb^Wc$$~~

~~$$W \rightarrow b$$~~

~~$$Z \rightarrow a$$~~

$$S \rightarrow X \mid Y \mid \epsilon$$

$$X \rightarrow aWc \mid aZc, \quad W \rightarrow bc$$

$$Y \rightarrow bWc \mid \quad Z \rightarrow ac$$

$$S \rightarrow aSc \mid X$$

$$X \rightarrow bXc \mid \epsilon$$

$$3) \phi$$

$$S \rightarrow S$$

(f)

~~$$S \rightarrow \{S\}$$~~

~~$$S \rightarrow [S]$$~~

~~$$S \rightarrow (S)$$~~

$$S \rightarrow [S] \mid \epsilon \mid SS$$

or

$$S \rightarrow S[S]S \mid \epsilon$$

7) $L = \{a^n b^m a^{2n} \mid n, m \geq 0\}$ 14/7/22

$$S \rightarrow aXa \mid \epsilon$$

$$X \rightarrow Ya \mid aYaa$$

$$Y \rightarrow Yb \mid \epsilon$$

$$S \rightarrow aSaa \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

8) All nonempty strings that start and end with same symbol

$$S \rightarrow X$$

$$X \rightarrow XSX$$

$$S \rightarrow SXs \mid sY$$

$$X \rightarrow Y \mid Za \mid Zb$$

$$Y \rightarrow \epsilon$$

$$S \rightarrow axa \mid bxb \mid a \mid b$$

$$X \rightarrow ax \mid bx \mid \epsilon$$

9) All strings with more a's than b's

$$S \rightarrow Aa \mid MS \mid SMA$$

$$A \rightarrow Aa \mid \epsilon$$

$$M \rightarrow \epsilon \mid MM \mid bMa \mid aMb$$

Derivations using CFG

(1)

$$S \rightarrow Aa$$

$$A \rightarrow Aa$$

$$A \rightarrow \epsilon$$

Derivation of $a^*(a+b00)$ by G_1

$$\begin{aligned}
 \rightarrow E &\Rightarrow E * E \Rightarrow E * (E) \Rightarrow E * (E + E) \Rightarrow E * (I + E) \\
 &\Rightarrow E * (a + E) \Rightarrow E * (a + I) \Rightarrow E * (a + I0) \\
 &\Rightarrow E * (a + I00) \Rightarrow E * (a + b00) \Rightarrow I * (a + b00) \\
 &\Rightarrow a * (a + b00)
 \end{aligned}$$

Rightmost Derivation

$$\begin{aligned}
 E &\Rightarrow E * E \Rightarrow E * (E) \Rightarrow E * (E + E) \Rightarrow E * (E + I) \Rightarrow E * (E + I0) \\
 &\Rightarrow E * (E + I00) \Rightarrow E * (E + b00) \Rightarrow E * (aI + b00) \\
 &\Rightarrow E * (a + b00) \Rightarrow I * (a + b00) \Rightarrow a * (a + b00)
 \end{aligned}$$

15/7/22 Parse tree or derivative tree :

$$S \rightarrow AB$$

$$A \rightarrow aAA$$

$$A \rightarrow a$$

$$B \rightarrow bB$$

$$B \rightarrow b$$

~~$G \neq \{$~~

~~$\}$~~

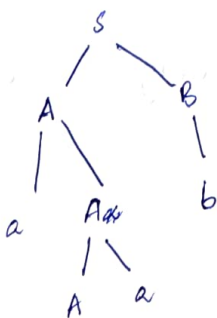
$\rightarrow$ Root can be any nonterminal

$\rightarrow$ leaf nodes can be terminals or nonterminals

$\rightarrow$ A derivation tree with root S shows the production used to obtain a sentential form.

Derivation of yield

$aAab$



In the previous example & ambiguity was resolved

by $E \rightarrow (E)$

(10) $E \rightarrow E * E$

$\rightarrow E + E$

$\rightarrow \epsilon$

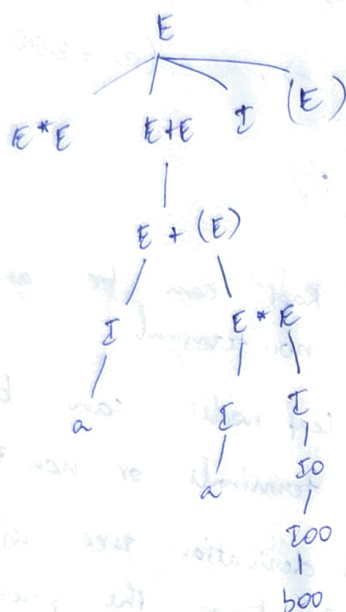
$\rightarrow (E)$

otherwise

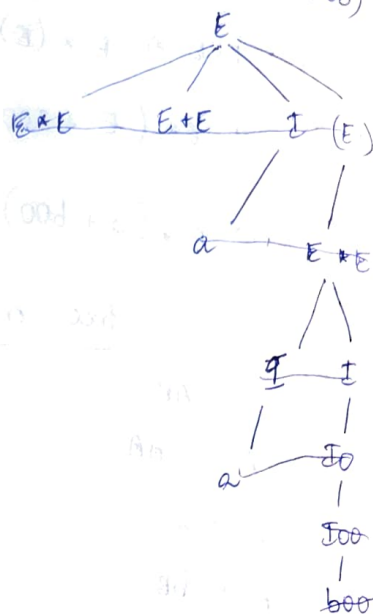
$E * E + E$

$\Rightarrow$ ambiguous.

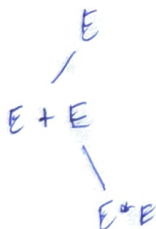
for $a + (a * b00)$



for $a + (a * b00)$



for $a + a * b00$



or



Two parse trees $\Rightarrow$ ambiguous grammar

Q Consider the following grammar

$$S \rightarrow SS \mid AAA \mid \epsilon$$

$$A \rightarrow aA \mid Aa \mid b$$

- ① Describe the language generated by this grammar,
 ② Get left-most derivation for abbaba

→ $L \rightarrow (a^*(a+b)^*.a^*)^*$

$$L = (a^*ba^*ba^*b)^*$$

$$= (a^*ba^*ba^*ba^*)^*$$

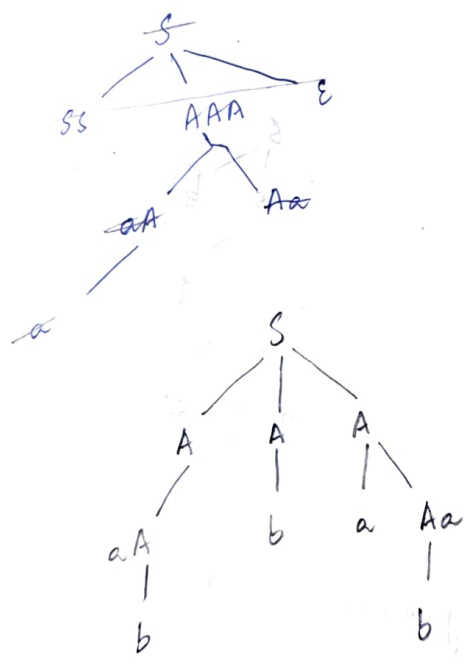
Left-most derivation for abbaba:

$$S \xrightarrow{lm} SS \xrightarrow{lm} AAA \xrightarrow{lm} AAAaAa$$

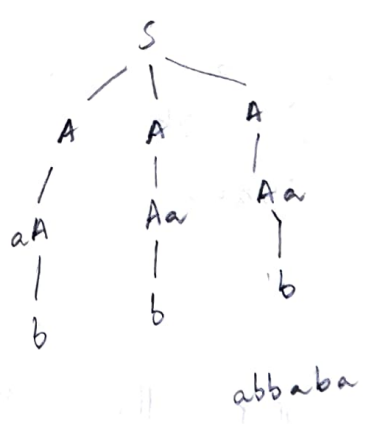
$$\xrightarrow{lm} aAAA \xrightarrow{lm} abAA$$

$$\xrightarrow{lm} abAaA \xrightarrow{lm} abbaA$$

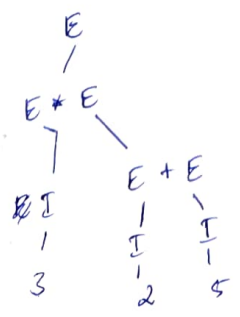
$$\xrightarrow{lm} abbaAa \xrightarrow{lm} abbaba$$



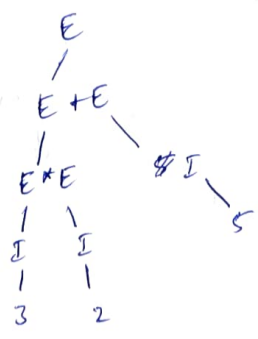
$S \rightarrow AAA \rightarrow abAA$
 $\rightarrow abbA \rightarrow abbaA \rightarrow abbaAa \rightarrow abbaba$



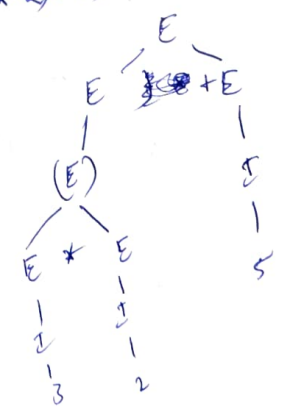
$3 * 2 + 5$



4



$(3 * 2) + 5$



Causes of Ambiguity:

- 1) Precedence of operators is not considered.
- 2) A sequence of same operators can be grouped from left to right eg: $E+E+E$ or $E * E * E$

To remove ambiguity in $E+E * E$ instead use
 $E \rightarrow E+E$
 $T \rightarrow T * T$

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$$L = \{ w^R w^R, w \in (a,b)^* \}$$

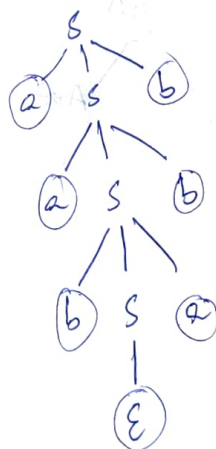
and w^R indicates is in which occurrences of 'a' is replaced by 'b' and vice-versa.

Give CFG that generates L and a parse tree that shows

$$\rightarrow aababbb \in L$$

$$S \rightarrow asb \mid bsa \mid \epsilon$$

$$\begin{array}{ll} asb & , s \rightarrow asb \\ \downarrow & \\ aasbb & , s \rightarrow asb \\ \downarrow & \\ aabbsabb & , s \rightarrow bsa \\ \downarrow & \\ aababbb & , s \rightarrow \epsilon \end{array}$$

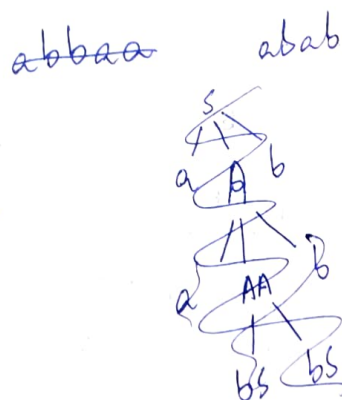
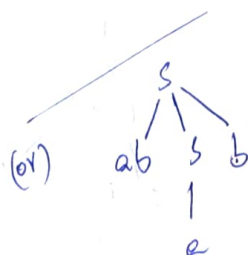


3) Show that the context free grammar with production rule

$$S \rightarrow a \mid aAb \mid abSb$$

$$A \rightarrow aAAb \mid bS$$

$$\begin{aligned} S &\rightarrow aAb \rightarrow abSb \\ &\rightarrow abab \end{aligned}$$



8)

$$S \rightarrow aSS$$

ababb

aababbbb

$$S \rightarrow b$$

$$\begin{aligned}
 S &\rightarrow aSS \\
 &\rightarrow aabSSS \\
 &\rightarrow \dots
 \end{aligned}$$

$$\begin{aligned}
 S &\rightarrow aSS \rightarrow aaSSS \\
 &\rightarrow aabSS \\
 &\rightarrow aabaSSS \\
 &\rightarrow aababbbb
 \end{aligned}$$

to solve ambiguity
 For $E + E + E$

Instead, use $E \rightarrow T + E \mid \epsilon T$

$$T \rightarrow 0 \mid 1 \mid 2 \mid \dots \mid 9 \mid \text{II TT}$$

$$10 + 20 + 30 + 40$$

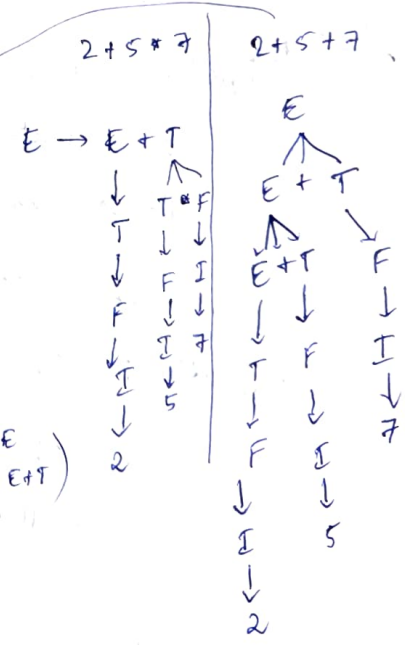
$$\begin{aligned}
 E &\rightarrow T + E \rightarrow TT + E \rightarrow T + T + T + E \rightarrow T + T + T + T \\
 &\rightarrow TT + TT + TT + TT \\
 &\rightarrow 10 + 20 + 30 + 40
 \end{aligned}$$

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$$\begin{aligned}
 I &\rightarrow a \mid b \mid Ia \mid Ib \mid IO \mid II \mid 0 \mid \dots \mid 9 \\
 F &\rightarrow I \mid (E) \\
 T &\rightarrow F \mid T * F \\
 E &\rightarrow T \mid E + T
 \end{aligned}$$

for to
 $a + b + b0$

(not ambiguous as we do not have $T + E$ in addition to $E + T$)



Simplification of CFG's

- * Eliminating useless symbols
- * Eliminating ϵ productions
- * unit

Useless Symbols:

- ↳ Non Terminating production
- ↳ Unreachable Symbols

eg: $S \rightarrow AB \mid C$
 $A \rightarrow aA \mid a$
 $B \rightarrow bB$
 $C \rightarrow c$

$\Rightarrow$ B removed

$S \rightarrow AC$
 $A \rightarrow aA$
 $C \rightarrow c$

$\Rightarrow$ A is unreachable
 $S \rightarrow C$
 $C \rightarrow c$

⑨ Eliminate useless symbols in the grammar having production rules,

$S \rightarrow AC \mid B$

$A \rightarrow a$

$C \rightarrow c \mid B$

$E \rightarrow aA$

$T = \{a, c, e\}$

$V = \{S, A, B, C, E\}$

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↳ Epsilon Productions

To eliminate ϵ -productions, we need to discover nullable variables i.e. $\lambda \mid A \Rightarrow^* \epsilon$

Basis: If there is a production $A \rightarrow \epsilon$, A is nullable

Induction: If there is a production $A \rightarrow \alpha$, & all symbols of α are nullable, A is nullable

eg: $S \rightarrow AB$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow bB \mid A$

Basis: A is nullable because $A \rightarrow \epsilon$

Induction: B , " " $B \rightarrow A$

$\Rightarrow S$ is nullable as, $S \rightarrow AB$

Eliminating ϵ -productions,

eg: $S \rightarrow ABC$, A

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid \epsilon$

$C \rightarrow \epsilon$

A, B, C & S are nullable

New grammar,

$S \rightarrow ABC \mid AB \mid AC \mid BC \mid A \mid B \mid C$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

$\rightarrow C$ is directly ϵ & $\therefore$ removed.

$\rightarrow$ Take ϵ products of A, B, C considered in S
ie from right side, one 'x', becoming ϵ

gives $AB \mid AC \mid BC$,

||ly 'xx' becoming ϵ gives A, B, C

Thus. $S \rightarrow ABC \mid AB \mid AC \mid BC \mid A \mid B \mid C$

$\rightarrow$ Next,

$A \rightarrow aA \mid \epsilon \Rightarrow aA \mid a$ ($\because aA$ with $A \rightarrow \epsilon$ gives a)

||ly $B \rightarrow bB \mid \epsilon \Rightarrow bB \mid b$

$\rightarrow$ This way ϵ is supported.

Theorem: If L is a CFL, then $L - \{\epsilon\}$
is a CFG with no ϵ -production.

① Eliminate all ϵ -productions from.

$$S \rightarrow OAD \mid IB \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

i) identify all nullable variables

ii) Eliminate ϵ -prodⁿ & rewrite grammar.

C, A, B, S

$$\rightarrow S \rightarrow OAD \mid IB \mid BB \mid OO \mid II \mid B$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

②

$$S \rightarrow ASA \mid aB \mid b$$

$$A \rightarrow B$$

$$B \rightarrow b \mid \epsilon$$

$\Rightarrow$

$$S \rightarrow ASA \mid aB \mid b \mid AS \mid SA \mid aS$$

$$A \rightarrow B$$

$$B \rightarrow b$$

A & B are nullable

Unit Productions: (rhs is a single non-terminal)

$$\text{eg: } \begin{array}{l} A \rightarrow B \\ B \rightarrow b \end{array} \Rightarrow \begin{array}{l} A \rightarrow b \\ B \rightarrow b \end{array}$$

ie if $A \rightarrow B$ & $B \rightarrow a$, then add productⁿ $A \rightarrow a$,
where $a \rightarrow$ non-unit production & then
drop all unit productions.

$$\text{eg: } \begin{array}{l} A \rightarrow B \\ B \rightarrow C \\ C \rightarrow D \end{array} \Rightarrow A \rightarrow D$$

8) Remove all unit productions from all following prodⁿ

$$S \rightarrow xy$$

$$x \rightarrow a$$

$$\Rightarrow S \rightarrow xy$$

$$y \rightarrow z/b$$

$$y \rightarrow a/b$$

$$z \rightarrow M$$

$$x \rightarrow a$$

$$M \rightarrow N$$

$$N \rightarrow a$$

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9) $S \rightarrow AB$

$$S \rightarrow AB/A$$

$$A \rightarrow a$$

$\Rightarrow$

$$A \rightarrow a$$

$$B \rightarrow CD$$

$$B \rightarrow CD/D/C$$

$$C \rightarrow aC/\epsilon$$

$$D \rightarrow bDa/ba$$

$$D \rightarrow bDa/\epsilon$$

$$C \rightarrow aC/a$$

Chomsky Normal Form

A CFG is said to be in CNF if

$$A \rightarrow BC \quad (\text{at least } 2 \text{ non-terminal})$$

$$A \rightarrow a \quad (1 \text{ terminal})$$

$A \rightarrow BCDE$ is replaced by

$$A \rightarrow BF$$

$$F \rightarrow CG$$

$$G \rightarrow DE$$

F & G must be used nowhere else

Algo to convert to CNF

1) If $S \rightarrow SA$, create new start symbol S' , i.e. $S' \rightarrow S$

2) Remove Null productⁿ

3) Remove unit productⁿ

4) Replace $A \rightarrow B_1 \dots B_n$ where $n \geq 2$ with $A \rightarrow B_1 C$, $C \rightarrow B_2 \dots B_n$

Repeat this step

5) If $A \rightarrow aB \Rightarrow A \rightarrow XB$, $X \rightarrow a$

8) Convert the following CFG into CNF

a) $S \rightarrow ASA \mid ab$

$A \rightarrow B \mid s$

$B \rightarrow b \mid \epsilon$

A & B are nullable

~~$S' \rightarrow SS$~~

$S' \rightarrow S$
 $S \rightarrow ASA \mid XB \mid BSB \mid SSS$

$X \rightarrow a$

~~$A \rightarrow s$~~

$B \rightarrow b$

$\Downarrow$

~~$S' \rightarrow S$~~

~~$S' \rightarrow XB \mid a \mid BS \mid \epsilon B \mid SS'$~~

~~$X \rightarrow a$~~

~~$B \rightarrow b$~~

i) $S' \rightarrow S$
 $S \rightarrow ASA \mid ab$
 $A \rightarrow B \mid s$
 $B \rightarrow b \mid \epsilon$

ii) $S' \rightarrow S$
 $S \rightarrow ASA \mid ab \mid s \mid a \mid SA \mid AS$
 ~~$A \rightarrow B \mid s$~~
 $B \rightarrow b$

iii) $S' \rightarrow ASA \mid ab \mid s \mid a \mid AS \mid SA$
 $A \rightarrow b \mid s$

$\Rightarrow$ $S' \rightarrow ASA \mid aA \mid s \mid a \mid AS \mid SA$
 $A \rightarrow b \mid s$
 $\begin{matrix} A \rightarrow s \\ S \rightarrow a \end{matrix}$

$\Downarrow$

$S' \rightarrow ASA \mid aA \mid s \mid a \mid AS \mid SA$
 $A \rightarrow a \mid b$

iv) ~~$S' \rightarrow AX \mid a \mid AS \mid SA$~~
 ~~$X \rightarrow SA$~~
 ~~$A \rightarrow AS$~~
 $A \rightarrow a \mid b$

$S' \rightarrow aA \mid a$
 $A \rightarrow a \mid b$

$\Rightarrow$

v) ~~$S' \rightarrow XA \mid a$~~
 ~~$X \rightarrow a$~~
 ~~$A \rightarrow a \mid b$~~

Answer :

$$S' \rightarrow AX | YB | a | AS | SA$$

$$S \rightarrow AX | YB | a | AS | SA$$

$$A \rightarrow bA | S \rightarrow b | AX | YB | a | AS | SA$$

$$B \rightarrow b$$

$$X \rightarrow SA$$

$$Y \rightarrow a$$

⑧ Convert to CNF form

$$A \rightarrow BAB | B | \epsilon$$

28/7/22 $B \rightarrow \epsilon$

i) $A' \rightarrow BAB | B | \epsilon$
 $A \rightarrow BAB | B | \epsilon$
 $B \rightarrow \epsilon$

ii)

$$S \rightarrow A' \rightarrow BAB | BA | AB | BB | \epsilon A | B$$

$$A \rightarrow BAB | BA | AB | BB | \epsilon A | B$$

$$B \rightarrow \epsilon$$

iii)

$$A' \rightarrow BAB | BA | AB | BB | \epsilon$$

$$A \rightarrow BAB | BA | AB | BB | \epsilon$$

$$B \rightarrow \epsilon$$

iv)

$$A' \rightarrow BC | BA | AB | BB | \epsilon DD$$

$$A \rightarrow BC | BA | AB | BB | \epsilon DD$$

$$C \rightarrow AB$$

$$B \rightarrow \epsilon DD$$

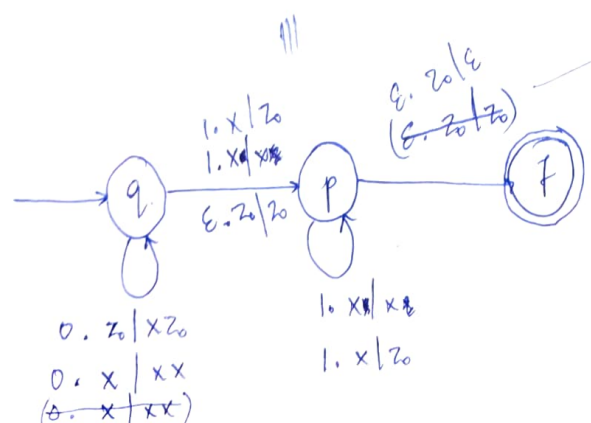
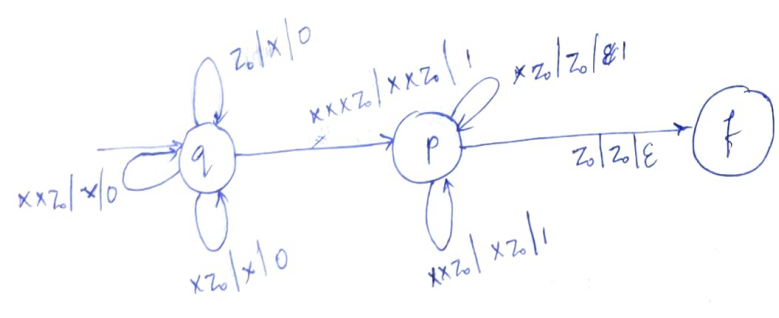
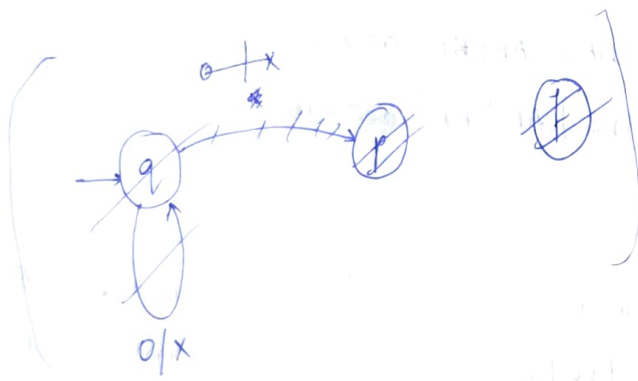
$$D \rightarrow \epsilon$$

PUSH DOWN AUTOMATA (PDA)

1/8/22 ID - Instantaneous Description : 5 tuple system
 ↳ Goes to → used to represent transition δ/w

eg: $(q_0, \underbrace{11611}_{i/p}, \#)$ start $\vdash (q_0, 1\epsilon11, L\#)$
 $\vdots$
 $(q_1, 6, \#) \vdash (q_2, 6, \epsilon)$

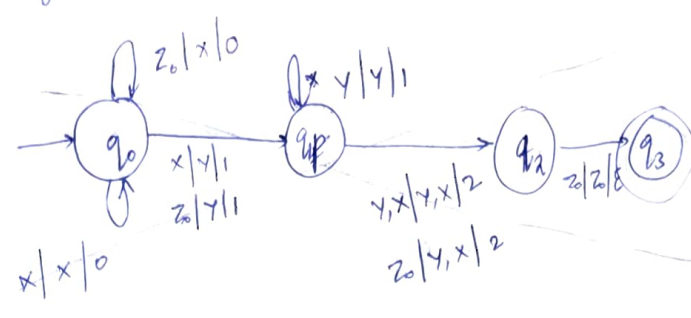
⑧ $(q, 000111, z_0) \vdash (q, 00111, xz_0) \vdash (q, 0111, xxz_0)$
 $\vdash (q, 111, xxxz_0) \vdash (p, 11, xxz_0) \vdash (p, 1, xz_0)$
 $\vdash (p, \epsilon, z_0) \vdash (f, \epsilon, z_0)$



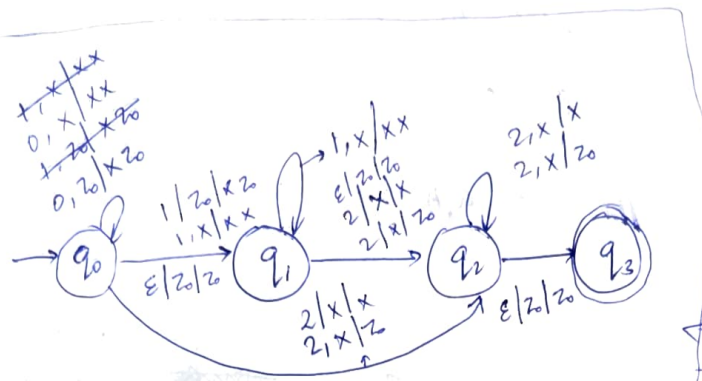
i/p - top of stack / Next top

4/8/22

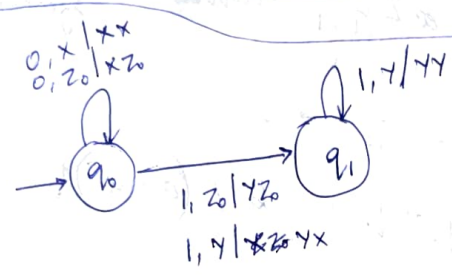
9) Design a PDA, for $0^m 1^n 2^{m+n}$ $m, n \geq 0$



PDA:
 $Q \rightarrow \{q_0, q_1, q_2, q_3\}$
 $\Sigma \rightarrow \{0, 1, 2\} \cup \epsilon$
 $P = \{x, z_0\}$



q_0
 $z_0 \rightarrow$ stack start symbol
 $f = \{q_3\}$



~~0 1 2~~

00 111 2222

$(q_0, 001112222, z_0) \vdash (q_0, 01112222, xz_0) \vdash (q_0, 01112222, xxz_0)$
 $\vdash (q_1, 112222, xxxz_0) \vdash (q_1, 12222, xxxxxz_0)$
 $\vdash (q_1, 2222, xxxxxz_0) \vdash (q_2, 222 xxxxxz_0)$
 $\vdash (q_2, 22, xxxz_0) \vdash (q_2, 2, xxz_0) \vdash (q_2, \epsilon, xz_0)$

Obtain CFG that generates the language accepted ^{SLN} by PDA

$$P = \{Q, \Sigma, \Gamma, q_0, Z_0, \delta, F\}$$

$$Q = \{q_0, q_1\}$$

$$F = \{q_1\}$$

$$\Sigma = \{a, b\}$$

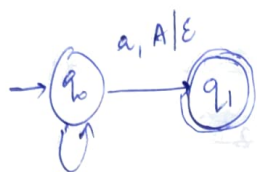
$$\delta\{q_0, q_1, Z\} = \{q_0, AZ\}$$

$$\Gamma = \{A, Z\}$$

$$\delta\{q_0, b, A\} = \{q_0, AA\}$$

$$q_0 = q_0$$

$$\delta\{q_0, a, A\} = \{q_0, \epsilon\}$$



$$a, Z / AZ$$

$$b, A / AA$$

$$q_0 Z q_1 \quad \text{start symbol}$$

$$q_0 Z q_0 \rightarrow a$$

$$q_0 A q_0 \rightarrow b / (q_0 A q_0) b$$

$$q_0 A q_1 \rightarrow a$$

$$q_0 Z q_1 \rightarrow (q_0 Z q_0)(q_0 A q_0)(q_0 A q_1)$$

GNF

Q Convert following grammar to PDA. Show the instantaneous description for the string "aabbba"

$$S \rightarrow aABC$$

$$A \rightarrow aB / a$$

$$B \rightarrow bA / b$$

$$C \rightarrow a$$

$$\delta(q_0, \epsilon, Z_0) = (q_1, SZ_0)$$

$$\delta(q_1, a, S) = (q_1, ABC)$$

$$\delta(q_1, a, A) = (q_1, B)$$

$$\delta(q_1, a, A) = (q_1, \epsilon)$$

$$\delta(q_1, b, B) = (q_1, A)$$

$$\delta(q_1, b, B) = (q_1, \epsilon)$$

$$\delta(q_1, a, C) = (q_1, \epsilon)$$

8/8/22 Using pumping lemma

Show that the language $L = \{a^n b^n c^n, n \geq 0\}$ is not Context free L.

→ let $z = uvwxy$ and $|z| = 3n \geq n$

split z into $uvwxy$ such that $|vwx| \leq n$

and $|vx| \geq 1$

$$u = a^{n-(j+k+l)} \quad v = a^j \quad w = a^k \quad x = a^i$$

$$y = b^n c^n$$

$$|vwx| = j+k+l < n \quad |vx| = j+l \geq 1$$

$$uv^iwx^iy$$

$$\text{For } i=2 \quad z = a^{n-(j+k+l) + 2j+k+2l} b^n c^n \notin L$$

Closure properties of CFL :

① Union : $S \rightarrow S_1 \mid S_2$

② Concatenation : $S \rightarrow S_1 S_2$

③ Substitution : g: $S \rightarrow OSO \mid |S| \mid \epsilon \Rightarrow S_0 S S_0 \mid S_1 S S_1 \mid \epsilon$
 $S_0 \rightarrow a S_0 b \mid ab$
 $S_1 \rightarrow aa \mid bb$

④ Kleen closure : g: $V = V_1 \cup \{S\}$ where $S \notin V_1$
 $P \rightarrow P_1 \cup \{S \rightarrow SS_1 \mid \epsilon\}$ $G_1 = \{V_1, T, P_1, S\}$

⑥ Reversal:

$$L = \{ a^n b^n \mid n \geq 0 \} \text{ with } P: S \rightarrow a S b \mid \epsilon$$

$$L^R = \{ b^n a^n \mid n \geq 0 \} \text{ with } P^R: S \rightarrow b S a \mid \epsilon$$

for $L = \{ a^i b^j c^{i+j} \mid i, j \geq 0 \}$

$$L^R = \{ c^{i+j} b^j a^i \mid i, j \geq 0 \}$$

$$S \rightarrow S_1$$

$$S_1 \rightarrow a S_1 c \mid S_2$$

$$S_2 \rightarrow b S_2 c \mid \epsilon$$

$$S_2 \rightarrow c S_2 a \mid \epsilon$$

$$S_1 \rightarrow c S_2 a \mid S_2$$

$$(\underline{S_2 \rightarrow S})$$

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⑦ Intersection:

CFL is not closed under intersection

$$L = L_1 \cap L_2 = a^n b^n c^n, n \geq 0$$

cannot keep track of 3
need 2 stack

$$L_1 = a^n b^n c^m, n \geq 0, m \geq 0$$

$$L_2 = a^m b^m c^m, n \geq 0, m \geq 0$$

⑧ Complement:

$$L_1 \cap L_2 = L_1^c \cup L_2^c$$

closed under union

but LHS not closed

$\Rightarrow$ complement is not closed.

consider $a^n b^n$
complement is

a's more than b's
&&

& a's less than b's

both need to be accommodated in
same CFL

if grammar cannot be
written $\Rightarrow$ not closed

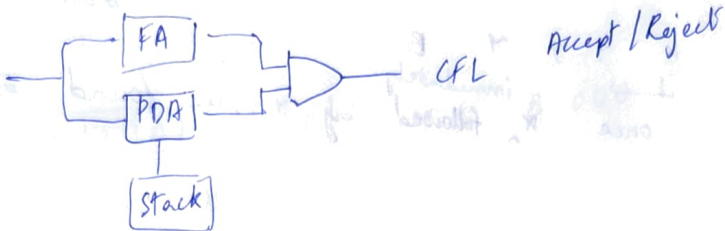
① Difference : CFL is not closed under difference

$$A - B = A \cap B'$$

$$L = \Sigma^* - L$$

a^*b^*
 a^nb^n

② Intersection with RL : Closed



Proof :

Let L_1 be a CFL and L_2 be a RL. We want to show that $L_1 \cap L_2$ is a CFL. We will use the Pumping Lemma for CFLs. Assume that $L_1 \cap L_2$ is not a CFL. Then there exists a string $w \in L_1 \cap L_2$ such that $w = uv^kx$ for some u, v, x and $k \geq 1$. Since $w \in L_1$, it must satisfy the Pumping Lemma for CFLs. Since $w \in L_2$, it must satisfy the Pumping Lemma for RLs. This leads to a contradiction, so $L_1 \cap L_2$ must be a CFL.

Let L_1 be a CFL and L_2 be a RL. We want to show that $L_1 \cap L_2$ is a CFL. We will use the Pumping Lemma for CFLs. Assume that $L_1 \cap L_2$ is not a CFL. Then there exists a string $w \in L_1 \cap L_2$ such that $w = uv^kx$ for some u, v, x and $k \geq 1$. Since $w \in L_1$, it must satisfy the Pumping Lemma for CFLs. Since $w \in L_2$, it must satisfy the Pumping Lemma for RLs. This leads to a contradiction, so $L_1 \cap L_2$ must be a CFL.

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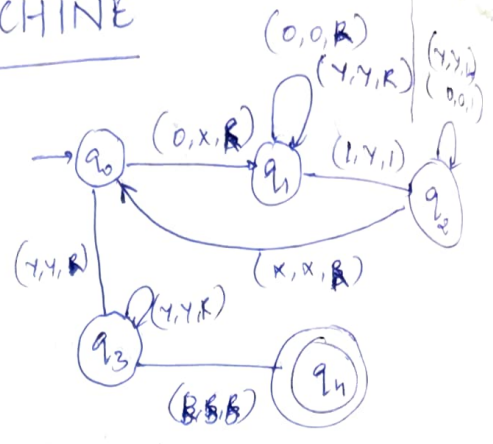
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TURING MACHINE

$$L = \{0^n 1^n, n \geq 1\}$$

0	0	0	1	1	1	B
x	0	0	y	1	1	B
x	x	0	y	y	1	B
x	x	x	y	y	y	B

once x_n immediately followed by y is found $\Rightarrow$ end.



$$0^n 1^n$$

(~~q₀ 00011B~~ + ~~x q₁ 011B~~)

- q₀ 0011B
- x q₁ 011B
- x 0 q₁ 11B
- x 0 y q₂ 1B
- x x q₁ y 1B
- x x y q₂ y B
- x x q₃ y y B
- x x y q₃ y B
- x x y y B q₄

$$0^n 1^n 2^n$$

- B 1001001B
- B q₀ 1001001B
- B y q₁ 001001B
- B y 001001 q₅ B
- B x 00100 q₃ y
- B y q₃ 00100 y B
- B y q₀ 00100 y B
- B y x q₁ 00100 y B
- B y x 0100 q₁ y B
- B y x 0100 q₂ y B
- B y x 010 q₃ x y B
- B y x q₃ 010 x y B
- B y x q₀ 010 x y B
- B y x x q₁ 010 x y B
- B y x x 010 q₂ x y B
- B y x x 01 q₃ x x y B
- B y x x 01 x x y B
- B y x x q₀ 01 x x y B
- B y x x q₄ x x y B
- B y x x y q₅ x x y B
- B y x x y x x y q₆ B

$$001122$$

- q₀ 001122 B
- x q₁ 01122 B
- x 0 q₁ 1122 B
- x 0 y q₂ 122 B
- x 0 y 1 q₂ 22 B
- x 0 y 1 z q₃ 2 B
- x 0 y 1 z 2 q₃ B
- x x q₁ y 1 z 2 q₃ B
- x x y q₂ z 2 q₃ B

~~x x y~~

- x q₃ 0 y 1 z 2 B
- x x q₁ y 1 z 2 B

$$x x y y z z q_4 B$$

results in
 $m-n$

if $m > n$ else 0

1e $3 \Rightarrow 111$, $2 \Rightarrow 11$

→ Blank on both sides, separation b/w 2 no.s given by 0

⇒

B	1	1	1	0	1	1	B
---	---	---	---	---	---	---	---

0Y 8 000 1 00 B

B B 1 1 0 1 B B

BBB 10 BBB

BBB BO BBB

BBB 10 BBB

⑧ Design

TM for checking palindrome over $\{0, 1\}$

B 011P B

B 0101 B

$$B \times I \times B$$

B x 31X 4 YB

$$B \times Y \times \underline{X \times B}$$
$$B \times Y \times Y \times B$$

B01103

$$B \times 11 \times B$$
$$B \times 71 \times B$$
$$B \times 478 B$$

6 1001 00 1 B

B 90 100 100 1 B

B 49₁₆ 001001₂

Y001001 25B

7001007 q_3^B

49001007 B

79.001007B

